



Mini Project Report – 17

**Bachelor of Computer Application
Semester – IV**

Sub: Full Stack Development

Topic: Crud Operations

By

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February-2026

REPORT ON CRUD OPERATIONS IN MONGODB (MDB)

1. Introduction to MongoDB (MDB)

MongoDB (MDB) is a NoSQL, document-oriented database designed to store and manage large volumes of unstructured and semi-structured data. Unlike traditional relational databases that store data in tables and rows, MongoDB stores data in flexible, JSON-like documents called BSON (Binary JavaScript Object Notation).

MongoDB is schema-less, which means it does not require a fixed structure for data storage. This flexibility allows developers to store different types of data in the same collection without predefined schemas. Due to its scalability, speed, and flexibility, MongoDB is widely used in modern web and mobile applications.

Key Features of MongoDB:

- Document-based storage (JSON-like format)
- Schema-less database structure
- High performance and scalability
- Support for indexing and aggregation
- Easy integration with programming languages

MongoDB is commonly used in applications such as e-commerce platforms, social media apps, real-time analytics systems, and cloud-based applications.

2. Requirements (MongoDB Community Server + MongoDB Compass)

To perform CRUD operations in MongoDB, the following software and system requirements are needed:

2.1 MongoDB Community Server

MongoDB Community Server is the core database engine used to store, manage, and retrieve data.

System Requirements:

- Operating System: Windows / Linux / macOS
- Minimum RAM: 4 GB (Recommended)
- Disk Space: At least 1 GB free
- Internet connection for downloading software
- Administrator privileges for installation

2.2 MongoDB Compass

MongoDB Compass is a graphical user interface (GUI) tool that allows users to visually interact with MongoDB databases without writing complex commands.

Advantages of MongoDB Compass:

- User-friendly interface
- Visual data management
- Easy CRUD operations
- Query building without coding
- Real-time data visualization

3. Difference Between MongoDB Community Server and MongoDB Compass

MongoDB Community Server and MongoDB Compass are two important components of the MongoDB ecosystem, but they serve different purposes.

3.1 MongoDB Community Server

MongoDB Community Server is the core database software that runs in the background and stores, manages, and processes the data. It is the actual database engine where all CRUD operations are executed.

Features of MongoDB Community Server:

- Stores data in collections and documents
- Executes database queries and commands
- Runs as a service on the system
- Supports high performance and scalability
- Command-line interaction using Mongo Shell

3.2 MongoDB Compass

MongoDB Compass is a graphical user interface (GUI) tool used to visually interact with MongoDB databases. It allows users to perform CRUD operations without writing commands.

Features of MongoDB Compass:

- Visual interface for database management
- Easy data insertion, update, and deletion
- Schema visualization
- Query builder for beginners
- Real-time data exploration

3.3 Key Differences Between Community Server and Compass

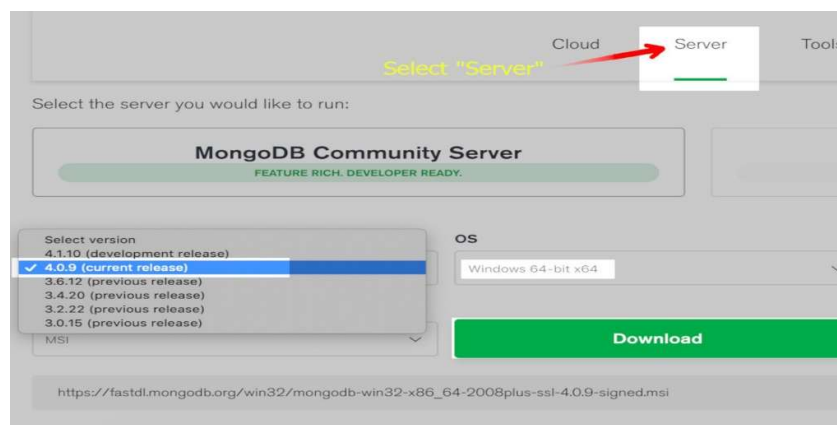
Feature	MongoDB Community Server	MongoDB Compass
Type	Database Engine	GUI Tool
Function	Stores and manages data	Visualizes and manages data
Interface	Command-line based	Graphical interface
Usage	Backend database operations	Frontend database management
Requirement	Mandatory to run database	Optional but recommended

3.4 Uses of Community Server and Compass

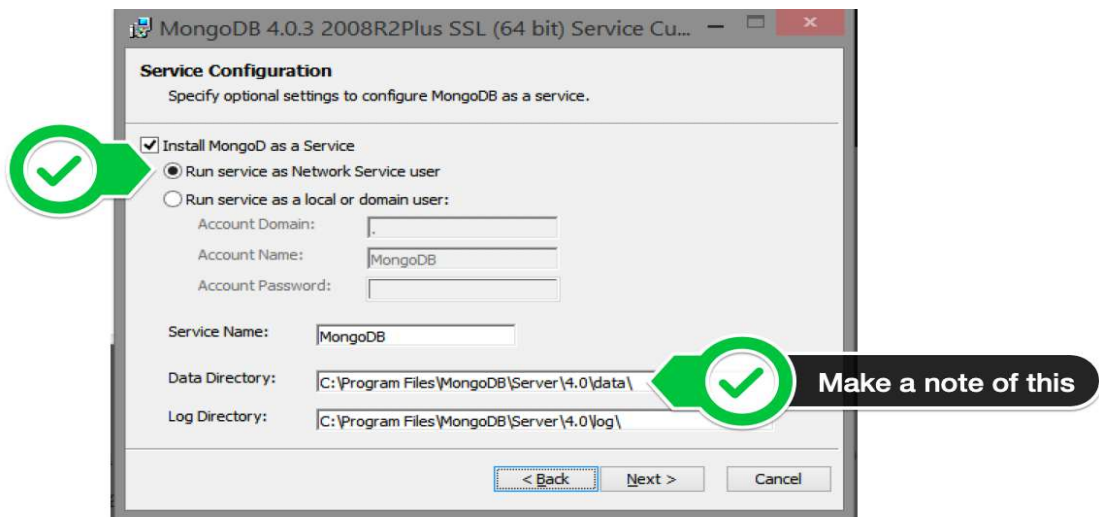
- Community Server is used to create and run databases and collections.
- Compass is used to visually perform CRUD operations and manage data easily.
- Together, they provide both backend processing and user-friendly database management.

4. Installation Steps of MongoDB and Compass

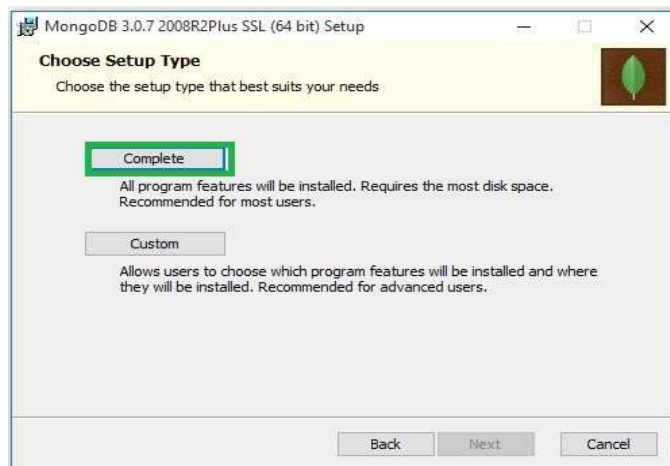
Step 1: Go to MongoDB Official Website & Download



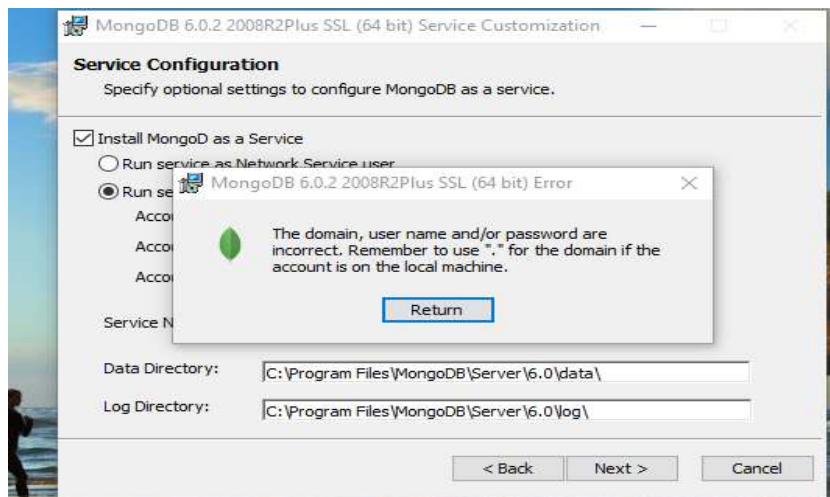
Step 2: Run the MongoDB Installer (.msi)



Step 3: Choose Setup Type (Complete Installation)

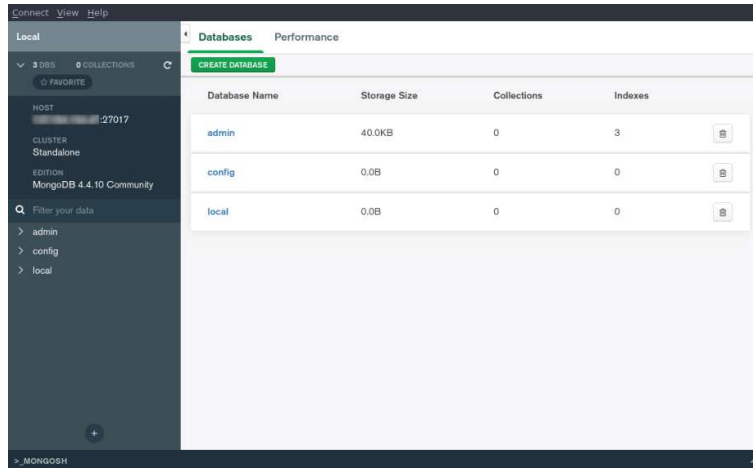


Step 4: Install MongoDB as a Service & Compass Option

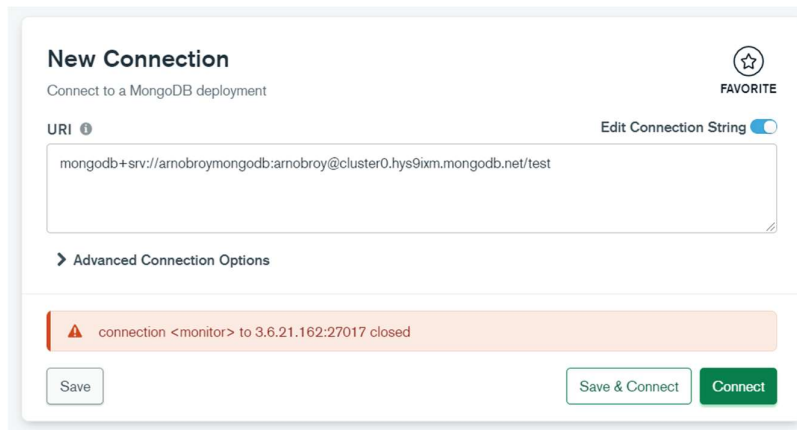


For Mongo DB compass

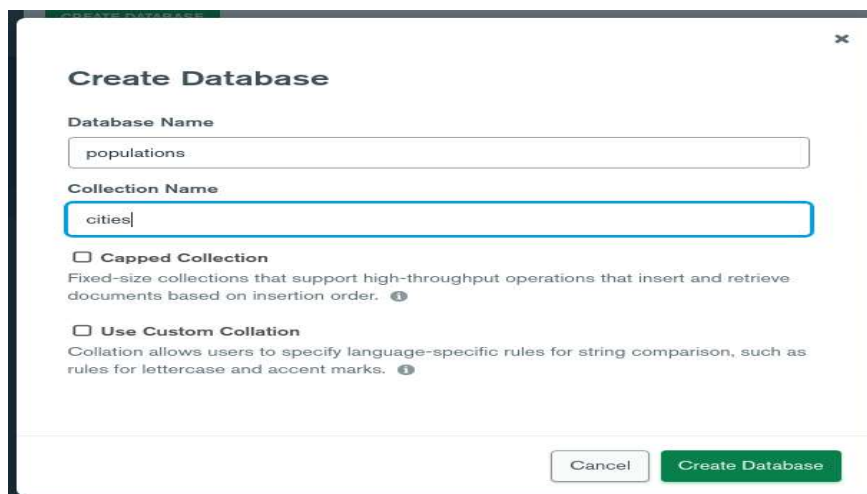
Step 1: Open MongoDB Compass



Step 2: Connect to Localhost



Step 3: Create a New Database



Step 4: Create a Collection

5. *CRUD Operations in MongoDB*

CRUD stands for Create, Read, Update, and Delete. These are the basic operations used to manage data in any database system.

5.1 Create Operation (Insert Data)

The Create operation is used to insert new documents into a collection.

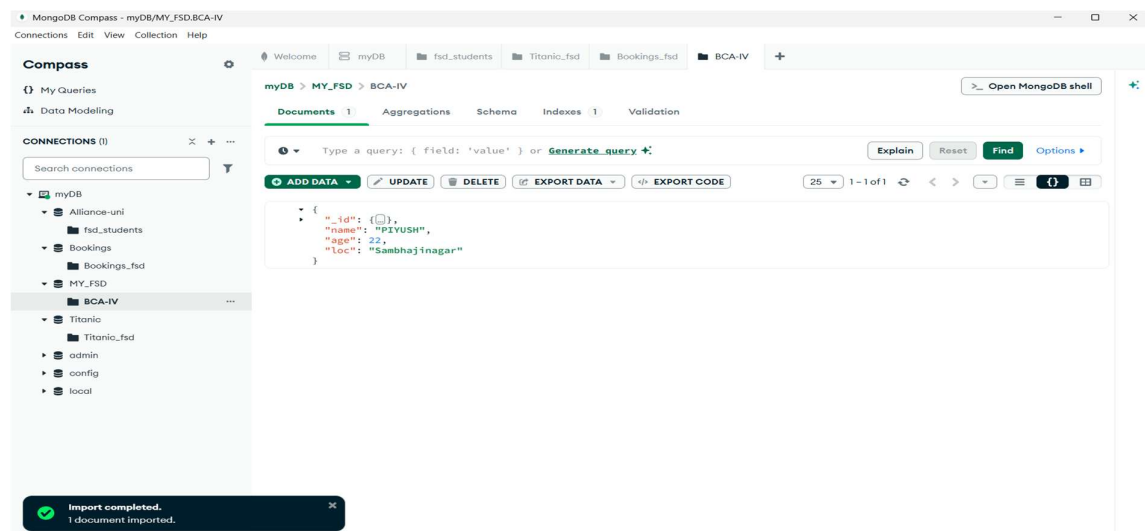
Step in Mongo Shell: use school

```
db.students.insertOne
```

```
(  
{ name: "John",  
  age: 20,  
  course: "Computer Science" }  
)
```

In MongoDB Compass:

1. Click Create Database
2. Database Name: school
3. Collection Name: students
4. Click Insert Document
5. Add JSON data
6. Click Insert



2.READ (Retrieve Data)

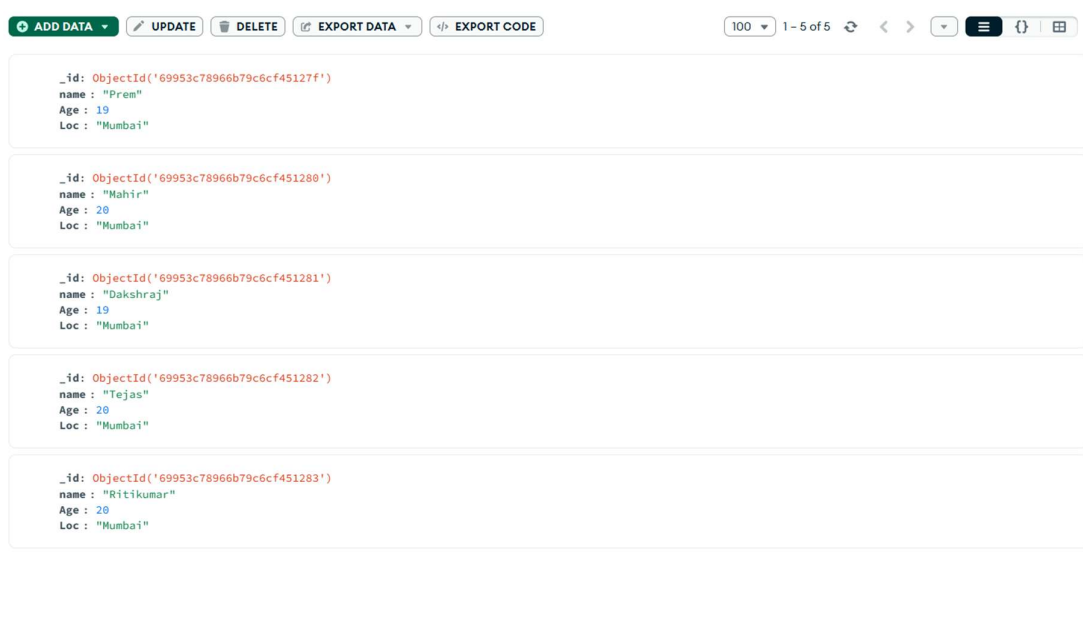
Find All Documents

```
db.students.find()
```

In Compass:

- Open collection
- Use Filter bar:

```
{ "name": "John" }
```



The screenshot shows the MongoDB Compass interface. At the top, there are buttons for 'ADD DATA', 'UPDATE', 'DELETE', 'EXPORT DATA', and 'EXPORT CODE'. Below these, a filter bar shows '100' and '1 - 5 of 5'. The main area displays a list of five documents, each with a red '_id' field and other fields like 'name', 'Age', and 'Loc'.

_id	name	Age	Loc
ObjectId('69953c78966b79c6cf45127f')	Prem	19	Mumbai
ObjectId('69953c78966b79c6cf451280')	Mahir	20	Mumbai
ObjectId('69953c78966b79c6cf451281')	Dakshraj	19	Mumbai
ObjectId('69953c78966b79c6cf451282')	Tejas	20	Mumbai
ObjectId('69953c78966b79c6cf451283')	Ritikumar	20	Mumbai

3.UPDATE (Modify Data)

Update One:

```
db.students.updateOne(  
  { name: "John" },  
  { $set: { age: 21 } }  
)
```

In Compass:

1. Click Edit Document
2. Modify value
3. Click Update

myDB > Alliance-uni > fsd_students Open MongoDB shell

Documents 5 Aggregations Schema Indexes 1 Validation

Type a query: { field: 'value' } or [Generate query](#)

ADD DATA UPDATE DELETE EXPORT DATA EXPORT CODE 100 1-5 of 5

★ fsd_students

	_id ObjectId	name String	Age Int32	Loc String
1	ObjectId('69953c78966b79c...	"Prem"	19	mumbai
Document modified. CANCEL UPDATE				
2	ObjectId('69953c78966b79c...	"Mahir"	20	"Mumbai"
3	ObjectId('69953c78966b79c...	"Dakshraj"	19	"Mumbai"
4	ObjectId('69953c78966b79c...	"Tejas"	20	"Mumbai"
5	ObjectId('69953c78966b79c...	"Ritikumar"	20	"Mumbai"

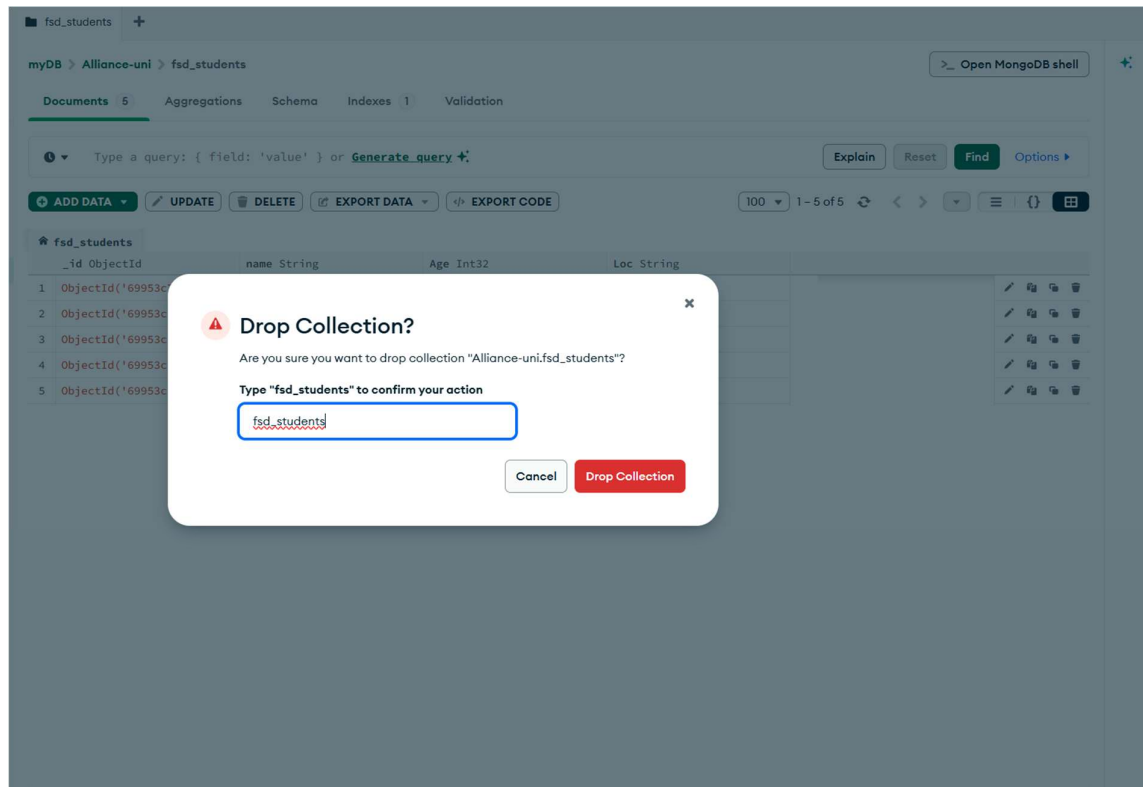
4.DELETE (Remove Data)

Delete One:

```
db.students.drop collection({ name: "John" })
```

In Compass:

1. Right Click
2. Select drop collection
3. Type document name
4. Click drop collection



6) Conclusion

In this mini project, we learned:

- Introduction to MongoDB (MDB)
- Installation of MongoDB & Compass
- Difference between MongoDB and SQL
- Complete CRUD Operations:
 - Create (Insert Data)
 - Read (View Data)
 - Update (Modify Data)
 - Delete (Remove Data)

MongoDB is simple, flexible, and very useful for modern applications, especially when handling large and unstructured data.